1 Expansions

We work with a family of bounded operators and with a step size that is smaller than one throughout.

Lemma 1. *The resolvent is bounded by a constant that does not depend on the step size.*

Proof. This is the usual estimate, and it needs no assumption beyond the ones already stated. □

Theorem 2. *The error has an expansion in powers of the step size.*

Proof. The argument continues by expanding each term and collecting the powers of the step size, which leaves a sum whose coefficients are bounded uniformly in the parameters. Every estimate below uses only the bound stated at the start of the proof, so the constants do not depend on the stiff part of the problem. Repeating this for the remaining terms gives the same conclusion, and the remainders are absorbed into the final constant.

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We substitute the first expansion into the second one and bound the resolvent. The terms collect as before, and we use Lemma 1:

$$\|x\| \le Ch.$$

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The remainder is of the size that the first estimate in this proof gives for $\|x\|, \|y\|, \|z\| \le Ch$ by Lemma 1(a), and the constant stays the same as before. The argument continues by expanding each term and collecting the powers of the step size, which leaves a sum whose coefficients are bounded uniformly in the parameters. Every estimate below uses only the bound stated at the start of the proof, so the constants do not depend on the stiff part of the problem. Repeating this for the remaining terms gives the same conclusion, and the remainders are absorbed into the final constant.

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Collecting everything gives the expansion. This completes the induction on the order of the expansion.
All done. □

The expansion has a first consequence, which the next statement records for later use in the following

Remark 1. The constants above are uniform in the step size.

Proposition 3. *The two errors agree to first order.*

Proof. Subtract the two expansions, which by the argument in Theorem 2 leaves the difference

$$e_1 - e_2 = \mathcal{O}(h^2).$$

□

The sentence after this proof does not belong to it, and it mentions Lemma 1.